

SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

The Abstract is a required part of your project. Bring 1 copy to Project Check-In.

Your abstract should be written after you finish your research and experimentation and should include:

- Project number & project title
- Full name of every team member
- Purpose of your project
- Procedure
- Observations/data/results
- Conclusions

Type or print neatly using 10- or 12- point black font, single spaced. Your abstract should be 250 words or less and fit within the frame.

(Project A59)

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ROCA: A Dynamic Metaheuristic Framework for Route Optimization Using Context-Aware Simulation of Ant Colony Systems

Travelling Salesman Problem (TSP) is a combinatorial optimization problem involving cyclic route planning, arising across a wide range of fields including emergency dispatch, networking, and computer-aided manufacturing. Due to superpolynomial time complexity, it is impractical to solve TSP via brute force. Over the last 40 years, many algorithms have been developed to find close-to-optimal solutions. However, the success of these heuristics is contingent on the assumption of a static solution space during computation. In real world applications of TSP, this assumption is often inadequate: traffic flow, latency, and available resources change continuously. This research presents a robust, environment-agnostic framework surrounding Ant Colony Optimization (ACO), a swarm-intelligence approach to TSP, designed to solve under dynamic costs and constraints. First, ACO's episodic structure cannot attribute specific choices to a "good" or "bad" solution. By introducing mid-cycle feedback, ROCA evaluates decisions at finer granularity. Second, metadata associated with each choice enables human language constraints to be injected directly into the algorithm's probabilistic selection step. Third, self-evolving parameters allow ROCA to converge more efficiently, independent of solution space uniformity. Compared to a contemporary ACO baseline implementation, ROCA reduced solution variability by an average of 24.58% and converged within 5% of newly introduced optima. Additionally, between 84.6-92.4% of contextual constraints were successfully incorporated. Evolutionary approaches outperformed their counterparts by roughly 4%. These improvements translate to faster emergency response times, saved compute, and accelerated production in time varying routing environments, laying groundwork for future domain-specific systems that fully bridge the theory-to-practice gap.